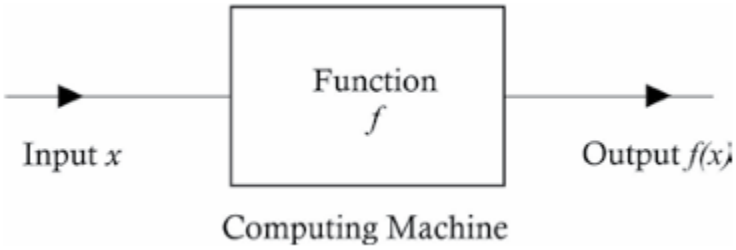
Functions and Types of Functions (Practice Sums from Lecture 4-6)

A function can be thought as a computing machine f that takes an input x, operates on it in some way, and produces exactly one output f(x). This output f(x) is called the value of f at x or image of x under f. The output f(x) is denoted by a single letter, say y, and we write y = f(x).



Example 1: Given $f(x) = x^3 - 2x^2 + 4x - 1$, find

(i) $f(0)$ (ii) $f(1)$ (iii) $f(-2)$ (iv) $f(1 + x)$

Solution: $f(x) = x^3 - 2x^2 + 4x - 1$

(i) $f(0) = 0 - 0 + 0 - 1 = -1$

(i) $f(1) = (1)^3 - 2(1)^2 + 4(1) - 1 = 1 - 2 + 4 - 1 = 2$

(ii) $f(-2) = (-2)^3 - 2(-2)^2 + 4(-2) - 1 = -8 - 8 - 8 - 1 = -25$

(iii) $f(1 + x) = (1 + x)^3 - 2(1 + x)^2 + 4(1 + x) - 1$

$= 1 + 3x + 3x^2 + x^3 - 2 - 4x - 2x^2 + 4 + 4x - 1$

$= x^3 + x^2 + 3x + 2$

Evaluate the function at the given arguments.

Q: 1 If $g(x) = (x^2 - 5x + 6)$, find:

(i) g(3)
(ii) g(-4)
(iii) g(2x)

Q: 2 If $h(t) = (4t^3 - t)$, find:

(i) h(1/2)
(ii) h(-2)
(iii) h(t - 1)

Q: 3 $f(x) = \frac{x^2-4}{x+3}$, find and simplify:

(i) f(-2)
(ii) f(a - 1)

Q: 4 $h(x) = \sqrt{x^2 + 5}$, find:

(i) h(2)
(ii) h(-3)

Key Point

The *domain* of a function is the set of possible inputs. The *range* of a function is the set of corresponding outputs.

Example 2: Let $f(x) = x^2$. Find the domain and range of f.

Solution: $f(x)$ is defined for every real number x.

Further for every real number x, $f(x) = x^2$ is a non-negative real number. So

Domain f = Set of all real numbers.

Range f = Set of all non-negative real numbers.

Example 3: Let $f(x) = \dfrac{x}{x^2 - 4}$. Find the domain and range of f.

Solution: At $x = 2$ and $x = -2$, $f(x) = \dfrac{x}{x^2 - 4}$ is not defined. So

Domain f = Set of all real numbers except -2 and 2 .

Range f = Set of all real numbers.

Q: 5 Find the domain and range of the following functions.

(i) $h(x) = x - 10$

(ii) $f(x) = 4 - 3x$

(iii) $g(x) = |x + 2|$

(iv) $f(x) = \sqrt{x - 4}$

(v) $f(x) = \dfrac{2x}{\sqrt{x^2 - 1}}$

Q: 6 A ball is thrown upward from the top of a building. The height h (in meters) after x seconds is approximately given by: $h(x) = 50 - 5x^2$

(i) Find the height of the ball when

(a) x=1 second

(b) x=2 seconds

(c) x=2.5 seconds

(ii) When does the ball hit the ground?

Q: 7 Draw the graphs of the following functions.

(i) The equation of line $y = -3x + 4$

(ii) The Parabola $y = x^2 - 4$

(iii) The circle $x^2 + y^2 = 25$